

Logic is reasoning.

Rules of Inference $\rightarrow$ independent of the discipline $\rightarrow$ Atomic sentence / proposition:

A declarative sentence which cannot be broken down into simpler sentences, these have values either true or false.

For example; (i) Ball is on the table. $\rightarrow$ (ii) Delhi is in Canada $\rightarrow F$ (iii) Give me a pen $\rightarrow$ Not a proposition.(iv) $1+101=111 \rightarrow F$ Let p, q and r be propositions.1. $\sim p \rightarrow$ Negation of p

p	$\sim p$
T	F
F	T

 p : Today is Monday. $\sim p$: Today is not Monday2. Disjunction ($\vee$)

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

 q : Clock is not working $p \vee q$: Today is Monday or clock is not working.3. Conjunction ($\wedge$)

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

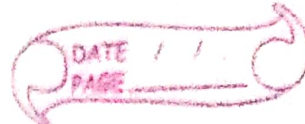
 $p \wedge q$: Today is Monday and clock is not working

Q.

Consider a proposition

p : You drive over 65 miles/hr.

q : You get a speeding ticket.



(a) You drive over 65 miles/hr and you get a speeding ticket.
 $p \wedge q$

(b) You don't drive over 65 miles/hr and you don't get a speeding ticket.
 $\neg p \wedge \neg q$

(c) Either you drive over 65 miles/hr or you get a speeding ticket.
 $p \vee q$

(d) Neither you drive over 65 miles/hr nor you get a speeding ticket.
 $\neg p \vee \neg q$

Q. Write in simple sentences if p and q are same as in the above exercises.

1. $\neg(p \wedge q)$

It is not true that you drive over 65 miles/hr and you get a speeding ticket.

2. $\neg p \vee q$

You don't drive over 65 miles/hr or you get a speeding ticket.

3. $\neg(\neg p \vee \neg q)$

It is not true that you don't drive over 65 miles/hr or you get a speeding ticket.

Q. Find truth table of the following proposition

1. $p \vee \sim q$

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p	q	$\sim q$	$p \vee \sim q$
T	T	F	T
T	F	T	T
F	T	F	F
F	F	T	T

2. $p \vee (\sim q \wedge p)$

p	q	$\sim q$	$\sim q \wedge p$	$p \vee (\sim q \wedge p)$
T	T	F	F	T
T	F	T	T	T
F	T	F	F	F
F	F	T	F	F

3. $p \wedge (q \vee \sim r)$

p	q	r	$\sim r$	$q \vee \sim r$	$p \wedge (q \vee \sim r)$
T	T	T	F	T	T
T	T	F	T	T	T
T	F	T	F	F	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	F
F	F	T	F	F	F
F	F	F	T	T	F

Q. Given p and q are true and r and s are false.
Find truth value of

(a) $(p \wedge r) \vee \neg [p \vee \neg q \wedge (r \vee s)]$

$\equiv (T \wedge F) \vee \neg [T \vee F \wedge (F \vee F)]$

$\equiv F \vee \neg [T \wedge F]$

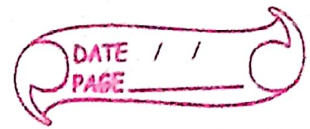
$\equiv F \vee \neg F \Leftrightarrow F \vee T \equiv T$ Ans



$$(b) (\neg(p \wedge q) \vee r) \vee ((\neg p \wedge q) \vee (\neg r) \wedge s)$$

$$\equiv (\neg(T \wedge T) \vee F) \vee ((F \wedge T) \vee T \wedge F)$$

$$\equiv F \vee T \equiv T \quad \text{Ans.}$$



→ Conditional $p \rightarrow q$ (if p then q)

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

→ Biconditional $(p \leftrightarrow q)$ p iff q

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Let p : You drive over 65 miles/hr.
 q : You get a speeding ticket.

Write the following proposition in symbolic form
 (a) You get a speeding ticket if you drive over 65 miles/hr.

If you drive over 65 miles/hr then you get a speeding ticket.

$$p \rightarrow q$$

(b) You get a speeding ticket if and only if you drive over 65 miles/hr.

$$q \leftrightarrow p$$

(c)

If you don't drive over 65 miles/hr then you will not get a speeding ticket.

$$\neg p \rightarrow \neg q$$

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If $p \rightarrow q$; then

Converse: $q \rightarrow p$

Inverse: $\neg p \rightarrow \neg q$

Contrapositive: $\neg q \rightarrow \neg p$

Inverse of contrapositive is converse.

Q. State the converse, inverse and contrapositive of the following.

(a)

If $\triangle ABC$ is a $\triangle$ then $|AB|^2 + |BC|^2 = |CA|^2$.

p : $\triangle ABC$ is a $\triangle$

q : $|AB|^2 + |BC|^2 = |CA|^2$

$$p \rightarrow q$$

Converse: $q \rightarrow p$

If $|AB|^2 + |BC|^2 = |CA|^2$ then $\triangle ABC$ is a $\triangle$.

Inverse: $\neg p \rightarrow \neg q$

If $\triangle ABC$ is not a $\triangle$ then $|AB|^2 + |BC|^2 \neq |CA|^2$.

Contrapositive: $\neg q \rightarrow \neg p$

If $|AB|^2 + |BC|^2 \neq |CA|^2$ then $\triangle ABC$ is not a $\triangle$.

(b)

If n is a prime then n is odd.

p : n is prime.

q : n is odd.

→ Converse: $q \rightarrow p$

If n is odd then n is a prime.

→ Inverse: $\neg p \rightarrow \neg q$
If n is not a prime then n is not odd.

→ Contrapositive: $\neg q \rightarrow \neg p$

If n is not odd then n is not prime.

(c) If n is a prime then n is odd or sum of even numbers.

p : n is prime

q : n is odd

r : n is sum of even numbers.

$$p \rightarrow (q \vee r)$$

Converse: $(q \vee r) \rightarrow p$

If n is odd or sum of even number then n is a prime.

Inverse: $\neg p \rightarrow \neg (q \vee r)$

If n is not a prime then it is not true that n is odd or sum of even numbers.

Contrapositive: $\neg (q \vee r) \rightarrow \neg p$

It is not the case that n is odd or sum of even number then n is not prime.

If $p, q \equiv T$ and $s \equiv F$.

Find truth value of;

(a) $(p \wedge \neg q) \rightarrow s$

$$\equiv (T \wedge F) \rightarrow F$$

$$\equiv F \rightarrow F \equiv T$$

(c) $(p \vee q) \leftrightarrow (p \rightarrow \neg s)$

$$\equiv (T \vee T) \leftrightarrow (T \rightarrow T)$$

$$\equiv T \leftrightarrow T$$

$$\equiv T$$

(b) $(p \leftrightarrow s) \rightarrow s$

$$\equiv (T \leftrightarrow F) \rightarrow F$$

$$\equiv F \rightarrow F \equiv T$$

→

Smallest Tautology

$$p \vee \neg p$$

→

Smallest Contradiction

$$p \wedge \neg p$$

Q. Which of the following are tautology and which are contradiction.

(a) $[p \wedge (p \rightarrow q)] \rightarrow p \rightarrow$ Tautology

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	$(p \wedge (p \rightarrow q)) \rightarrow p$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

(b) $\neg(q \rightarrow p) \wedge p \rightarrow$ Contradiction

p	q	$q \rightarrow p$	$\neg(q \rightarrow p)$	$\neg(q \rightarrow p) \wedge p$
T	T	T	F	F
T	F	T	F	F
F	T	F	T	F
F	F	T	F	F

→

Idempotent Law

(i)

$$p \vee p \equiv p$$

(ii)

$$p \wedge p \equiv p$$

→

Associative Law

(i)

$$p \vee (q \vee r) \equiv (p \vee q) \vee r$$

(ii)

$$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$$

→

Commutative Law

(i)

$$p \vee q \equiv q \vee p$$

(ii)

$$p \wedge q \equiv q \wedge p$$

→ Distributive Law

$$1. \quad p \vee (q \wedge r) \\ \equiv (p \vee q) \wedge (p \vee r)$$

$$2. \quad p \wedge (q \vee r) \\ \equiv (p \wedge q) \vee (p \wedge r)$$

→ Identity Law

$$1. \quad p \vee T \equiv T \\ p \vee F \equiv p$$

$$2. \quad p \wedge T \equiv p \\ p \wedge F \equiv F$$

$$\# \quad p \rightarrow q \equiv \neg p \vee q$$

$$\# \quad p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

De-Morgan's Law

$$1. \quad \neg(p \vee q) \equiv \neg p \wedge \neg q$$

$$2. \quad \neg(p \wedge q) \equiv \neg p \vee \neg q$$

Show the following equivalences.

$$(a) \quad (p \rightarrow q) \wedge (r \rightarrow q) \Leftrightarrow (p \vee r) \rightarrow q$$

Taking L.H.S

$$\equiv (p \rightarrow q) \wedge (r \rightarrow q)$$

$$\equiv (\neg p \vee q) \wedge (\neg r \vee q)$$

$$\equiv (\neg p \wedge \neg r) \vee q$$

$$\equiv \neg(p \vee r) \vee q$$

$$\equiv p \vee r \rightarrow q$$

$$\equiv \text{R.H.S}$$

(Using $p \rightarrow q \equiv \neg p \vee q$)

(Using distributive property)

($\neg p \vee q \equiv p \rightarrow q$)

$$(b) \quad p \rightarrow (q \rightarrow p) \Leftrightarrow \neg p \rightarrow (p \rightarrow q)$$

Taking L.H.S

$$\equiv p \rightarrow (q \rightarrow p)$$

$$\equiv p \rightarrow (\neg q \vee p)$$

$$\equiv \neg p \vee (\neg q \vee p)$$

$$\equiv (\neg p \vee p) \vee \neg q$$

$$\equiv T \vee \neg q$$

$$\equiv T$$

(Using $q \rightarrow p \equiv \neg q \vee p$)

(Using $p \rightarrow q \equiv \neg p \vee q$)

(Associative property)

$$\equiv T \vee q$$

$$\equiv (\neg p \vee p) \vee q$$

$$\equiv p \vee (\neg p \vee q)$$

$$\equiv p \vee (p \rightarrow q)$$

$$\equiv \neg p \rightarrow (p \rightarrow q)$$

$$\equiv \text{R.O.S}$$

$$(\neg p \vee p \equiv T)$$

(Associative property)

$$(\neg p \vee q \equiv p \rightarrow q)$$

$$(p \rightarrow q \equiv \neg p \vee q)$$

(c)

$$(q \vee r) \rightarrow r \equiv q \rightarrow r$$

Taking L.H.S

$$\equiv (q \vee r) \rightarrow r$$

$$\equiv \neg (q \vee r) \vee r$$

$$(p \rightarrow q \equiv \neg p \vee q)$$

$$\equiv (\neg q \wedge \neg r) \vee r$$

(De-Morgan's Law)

$$\equiv (\neg q \vee r) \wedge (\neg r \vee r)$$

(Distributive property)

$$\equiv (\neg q \vee r) \wedge T$$

$$(\neg r \vee r \equiv T)$$

$$\equiv (\neg q \vee r)$$

$$(p \wedge T \equiv p)$$

$$\equiv q \rightarrow r$$

$$(\neg p \vee q \equiv p \rightarrow q)$$

$$\equiv \text{R.O.S}$$

Ans.

Simplify

(d)

$$[(p \rightarrow q) \wedge p] \rightarrow q$$

Taking

$$\equiv [(p \rightarrow q) \wedge p] \rightarrow q$$

$$\equiv \neg [(p \rightarrow q) \wedge p] \vee q$$

$$(p \rightarrow q \equiv \neg p \vee q)$$

$$\equiv \neg ((\neg p \vee q) \wedge p) \vee q$$

$$\equiv \neg ((\neg p \wedge p) \vee (q \wedge p)) \vee q$$

(Using distributive property)

$$\equiv \neg (F \vee (q \wedge p)) \vee q$$

$$(\neg p \wedge p \equiv F)$$

$$\equiv \neg (q \wedge p) \vee q$$

$$(p \vee F \equiv p)$$

$$\equiv (\neg q \vee \neg p) \vee q$$

(De-Morgan's Law)

$$\equiv (\neg q \vee q) \vee \neg p$$

(Associative property)

$$\equiv T \vee \neg p$$

$$(T \vee p \equiv T)$$

$$\equiv T$$

Ans.

(e)

$$\neg (p \leftrightarrow q) \equiv (p \wedge q) \wedge [\neg (p \wedge q)]$$

Taking L.H.S

$$\equiv \neg (p \leftrightarrow q)$$

$$\equiv \neg ((p \rightarrow q) \wedge (q \rightarrow p)) \quad \{ p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p) \}$$

$$\equiv \neg ((\neg p \vee q) \wedge (\neg q \vee p)) \quad \{ p \rightarrow q \equiv \neg p \vee q \}$$

$$\equiv \text{Let } \neg p \vee q \equiv \alpha$$

$$\therefore \equiv \neg (\alpha \wedge (\neg q \vee p))$$

$$\equiv \neg ((\alpha \wedge \neg q) \vee (\alpha \wedge p)) \quad (\text{Distributive property})$$

$$\equiv \neg ((\neg p \vee q) \wedge \neg q) \vee ((\neg p \vee q) \wedge p)$$

$$\equiv \neg ((\neg p \vee (q \wedge \neg q)) \vee ((\neg p \wedge p) \vee q)) \quad (\text{Associative})$$

$$\equiv \neg (F \vee F) \vee (F \vee q)$$

$$\equiv \neg F \vee q$$

$$\rightarrow \text{Distributive}$$

$$\equiv \neg ((\neg p \wedge \neg q) \vee (q \wedge \neg q) \vee ((\neg p \wedge p) \vee (q \wedge p)))$$

$$\equiv \neg ((\neg p \wedge \neg q) \vee F \vee (F \vee (q \wedge p)))$$

$$\equiv \neg ((\neg p \wedge \neg q) \vee (q \wedge p)) \quad \left\{ \begin{array}{l} q \wedge \neg q \equiv F \\ p \wedge \neg p \equiv F \end{array} \right\}$$

$$\equiv \neg ((\neg p \wedge \neg q) \vee (q \wedge p)) \quad \{ p \vee F \equiv p \}$$

$$\equiv (p \vee q) \wedge \neg (q \wedge p) \quad (\text{De-Morgan's Law})$$

$$\equiv (p \vee q) \wedge \neg (p \wedge q) \quad (\text{Associative Law})$$

$$\equiv \text{R.H.S}$$

$$\underline{Q.} \quad (p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q) \equiv (p \vee q)$$

$$\therefore \equiv ((p \wedge q) \vee (\neg p \wedge q)) \vee (p \wedge \neg q) \quad (\text{Associative Law})$$

$$\equiv ((p \vee \neg p) \wedge q) \vee (p \wedge \neg q) \quad (\text{Distributive Law})$$

$$\equiv (T \wedge q) \vee (p \wedge \neg q)$$

$$\equiv q \vee (p \wedge \neg q) \quad (T \wedge q \equiv q)$$

$$\equiv (q \vee p) \wedge (q \vee \neg q) \quad (\text{Distributive Law})$$

$$\equiv (q \vee p) \wedge T \quad (q \vee \neg q \equiv T)$$

$$\equiv (q \vee p) \quad (p \wedge T \equiv p)$$

$$\equiv (p \vee q) \quad (\text{Associative Law})$$

$$\equiv \text{R.H.S}$$

→ Tautological implication

1. Take B is false and prove A is false.
2. Take A is true and prove B is true.

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Q Show the following implications without constructing the truth table.

(a) $P \rightarrow Q \Rightarrow P \rightarrow (P \wedge Q)$

Let $P \rightarrow (P \wedge Q)$ be false

$\therefore$ This can only hold if

$$P \equiv T$$

$$\text{and } P \wedge Q \equiv F$$

$$\Rightarrow Q \equiv F$$

$$\text{Hence, } P \rightarrow Q \equiv F \quad \text{L.H.S} = \text{R.H.S}$$

(b) $(P \rightarrow Q) \rightarrow Q \Rightarrow P \vee Q$

Let $P \vee Q$ be false

$\therefore$ This can only hold if

$$P \equiv F \text{ and } Q \equiv F$$

$$\therefore (P \rightarrow Q) \rightarrow Q$$

$$(F \rightarrow F) \rightarrow F$$

$$T \rightarrow F \equiv F$$

$$\text{Hence, } \text{L.H.S} = \text{R.H.S}$$

(c) $((P \vee \neg P) \rightarrow Q) \rightarrow ((P \vee \neg P) \rightarrow R) \Rightarrow Q \rightarrow R$

Let $Q \rightarrow R$ be false

$\Rightarrow$ This can only hold if

$$Q \equiv T \text{ and } R \equiv F$$

$$\text{As, } P \vee \neg P \equiv T$$

$$\therefore \equiv (T \rightarrow Q) \rightarrow (T \rightarrow R)$$

$$\equiv (T \rightarrow T) \rightarrow (T \rightarrow F)$$

$$T \rightarrow F \equiv F$$

$$\text{Hence, } \text{L.H.S} = \text{R.H.S}$$

Principle Disjunctive Normal Form (Sum of elementary products)

Obtain principle disjunctive Normal form of

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(i)

$$\rightarrow (p \rightarrow (q \wedge r))$$

p	q	r	$q \wedge r$	$p \rightarrow (q \wedge r)$	$\rightarrow (p \rightarrow (q \wedge r))$
T	T	T	T	T	F
T	T	F	F	F	$(T) \rightarrow p \wedge q \wedge \neg r$
T	F	T	F	F	$(T) \rightarrow p \wedge \neg q \wedge r$
T	F	F	F	F	$(T) \rightarrow p \wedge \neg q \wedge \neg r$
F	T	T	T	T	F
F	T	F	F	T	F
F	F	T	F	T	F
F	F	F	F	T	F

$$\therefore \text{P.D.N.F} \equiv (p \wedge \neg q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge \neg r)$$

(ii)

$$(\neg p \rightarrow r) \wedge (p \leftrightarrow q)$$

p	q	r	$\neg p$	$\neg p \rightarrow r$	$p \leftrightarrow q$	$(\neg p \rightarrow r) \wedge (p \leftrightarrow q)$
T	T	T	F	T	T	$T \rightarrow p \wedge q \wedge r$
T	T	F	F	T	T	$T \rightarrow p \wedge q \wedge \neg r$
T	F	T	F	T	F	F
T	F	F	F	T	F	F
F	T	T	T	T	F	F
F	T	F	T	F	F	F
F	F	T	T	T	T	$T \rightarrow \neg p \wedge \neg q \wedge r$
F	F	F	T	F	T	F

$$\therefore \text{PDNF} \equiv (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r)$$

(c)

$$p \wedge (\neg(q \wedge r)) \vee (p \rightarrow q)$$

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	p	q	r	$p \rightarrow q$	$q \wedge r$	$\neg(q \wedge r)$	$p \wedge (\neg(q \wedge r))$	$p \wedge (\neg(q \wedge r)) \vee (p \rightarrow q)$
✓	T	T	T	T	T	F	F	T
✓	T	T	F	T	F	T	T	T
✓	T	F	T	F	F	T	T	T
	T	F	F	F	F	T	T	T
✓	F	T	T	T	T	F	F	T
	F	T	F	T	F	T	F	T
	F	F	T	T	F	T	F	T
	F	F	F	T	F	T	F	T

As, the given expression is tautology, PDNF will contain every term.

$$\therefore \text{PDNF} = (p \wedge q \wedge r) \wedge (p \wedge q \wedge \neg r) \wedge (p \wedge \neg q \wedge r) \wedge (\neg p \wedge q \wedge r) \wedge (\neg p \wedge q \wedge \neg r) \wedge (p \wedge \neg q \wedge \neg r) \wedge (\neg p \wedge \neg q \wedge r) \wedge (\neg p \wedge \neg q \wedge \neg r)$$

Ans.

(d)

$$(p \vee q) \wedge (\neg p \wedge \neg q)$$

$$(p \vee q) \wedge (\neg(p \vee q)) \equiv F$$

Hence, PDNF does not exist.

□

To find PDNF without using truth table

Step 1: Remove all conditional and biconditional by $\vee, \wedge, \neg$ by this only.

Step 2: Eliminate negation before sums and products using de-Morgan's Law.

Step 3: Apply distributive property.

Step 4: Drop the terms which are contradiction

Step 5: If 'r' is missing in elementary product 'x' then take the product of x with tautology of r i.e.

$$x \wedge (r \vee \neg r) \equiv (x \wedge r) \vee (x \wedge \neg r)$$

Step 6: Delete identical minterms.

Q.

$$p \vee (p \wedge \neg q \wedge \neg r)$$

$$\therefore p \equiv p \wedge (q \vee \neg q) \equiv (p \wedge q) \vee (p \wedge \neg q) \text{ --- (1)}$$

Using Distributive property.

$\Rightarrow$

$$(p \wedge q) \equiv (p \wedge q) \wedge (r \vee \neg r) \equiv (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \text{ --- (2)}$$

and

$$(p \wedge \neg q) \equiv (p \wedge \neg q) \wedge (r \vee \neg r) \equiv (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \text{ --- (3)}$$

From 1, 2 and 3

$\therefore$

$$\text{PDNF: } (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \text{ Ans.}$$

Q.

$$(p \vee q) \wedge (\neg p \wedge \neg q)$$

$$\therefore (p \vee q) \wedge (\neg(p \vee q)) \equiv F$$

Hence, PDNF do not exist.

Q.

$$(\neg p \wedge q) \vee (p \wedge \neg r)$$

$\therefore$

$$\neg p \wedge q \equiv (\neg p \wedge q) \wedge (r \vee \neg r) \equiv (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \text{ --- (1)}$$

and

$$p \wedge \neg r \equiv (p \wedge \neg r) \wedge (q \vee \neg q) \equiv (p \wedge \neg r \wedge q) \vee (p \wedge \neg r \wedge \neg q) \text{ --- (2)}$$

From 1, 2

$$\text{PDNF: } (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (p \wedge \neg r \wedge q) \vee (p \wedge \neg r \wedge \neg q) \text{ Ans.}$$

$$Q. \quad \neg(p \rightarrow (q \wedge r))$$

$$\therefore \neg(\neg p \vee (q \wedge r)) \equiv p \wedge \neg(q \wedge r) \equiv p \wedge (\neg q \vee \neg r) \\ \equiv (p \wedge \neg q) \vee (p \wedge \neg r)$$

$$\therefore (p \wedge \neg q) \equiv (p \wedge \neg q) \wedge (r \vee \neg r) \equiv (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \quad - (1)$$

$$\text{and } (p \wedge r) \wedge (q \vee \neg q) \equiv (p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r) \quad - (2)$$

$\therefore$ From 1 and 2

$$PDNF: (p \wedge \neg q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge \neg r) \quad \text{Ans}$$

$$Q. \quad p \wedge [\neg(q \wedge r)] \vee (p \rightarrow q) \\ p \wedge (\neg q \vee \neg r) \vee (\neg p \vee q) \\ (p \wedge \neg q) \vee (p \wedge \neg r) \vee (\neg p \vee q)$$

$$\therefore p \wedge \neg q \equiv (p \wedge \neg q) \wedge (r \vee \neg r) \equiv (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \quad - 1 \\ p \wedge \neg r \equiv (p \wedge \neg r) \wedge (q \vee \neg q) \equiv (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge \neg r) \quad - 2$$

$$\neg p \equiv \neg p \wedge (q \vee \neg q) \equiv (\neg p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg p \wedge q \equiv (\neg p \wedge q) \wedge (r \vee \neg r) \equiv (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \quad - 3$$

$$\neg p \wedge \neg q \equiv (\neg p \wedge \neg q) \wedge (r \vee \neg r) \equiv (\neg p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r) \quad - 4$$

$$q \equiv q \wedge (p \vee \neg p) \equiv (p \wedge q) \vee (\neg p \wedge q)$$

$$p \wedge q \equiv (p \wedge q) \wedge (r \vee \neg r) \equiv (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \quad - (5)$$

From 1, 2, 3, 4 and 5
 PDNF: $(p \wedge q \wedge r) \vee (\neg p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg q \wedge r) \vee$
 $(p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge \neg r) \wedge (\neg p \wedge \neg q \wedge \neg r)$

Ans.

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Principle Conjunctive Normal Form (PCNF)
 (Product of Max Terms)

Q. Obtain PCNF of
 (i) $\neg(p \rightarrow (p \wedge r))$

p	q	r	$p \wedge r$	$p \rightarrow (p \wedge r)$	$\neg(p \rightarrow (p \wedge r))$
T	T	T	T	T	$F \rightarrow \neg p \vee \neg q \vee \neg r$
T	T	F	F	F	T
T	F	T	T	T	$F \rightarrow \neg p \vee \neg q \vee \neg r$
T	F	F	F	F	T
F	T	T	F	T	$F \rightarrow p \vee \neg q \vee \neg r$
F	T	F	F	T	$F \rightarrow p \vee \neg q \vee \neg r$
F	F	T	F	T	$F \rightarrow p \vee \neg q \vee \neg r$
F	F	F	F	T	$F \rightarrow p \vee \neg q \vee \neg r$

PCNF: $(\neg p \vee \neg q \vee \neg r) \wedge (\neg p \vee q \vee \neg r) \wedge (p \vee \neg q \vee \neg r) \wedge (p \vee \neg q \vee r) \wedge$
 $(p \vee q \vee \neg r) \wedge (p \vee q \vee r)$

(ii) $(p \wedge q) \wedge (\neg p \vee \neg q)$

p	q	$(p \wedge q) \wedge (\neg p \vee \neg q)$
T	T	$F \rightarrow \neg p \vee \neg q$
T	F	$F \rightarrow \neg p \vee \neg q$
F	T	$F \rightarrow p \vee \neg q$
F	F	$F \rightarrow p \vee \neg q$

PCNF = $(p \vee q) \wedge (p \vee \neg q) \wedge (\neg p \vee q) \wedge (\neg p \vee \neg q)$

Q.

$$p \wedge (p \vee \neg q \vee r)$$

$$(p \wedge p) \vee (p \wedge \neg q \vee r)$$

on with contradiction

$$p \equiv p \vee (q \wedge \neg q)$$

$$p \equiv (p \vee q) \wedge (p \vee \neg q)$$

$$(p \vee q) \equiv (p \vee q) \vee (r \wedge \neg r) \equiv (p \vee q \vee r) \wedge (p \vee q \vee \neg r)$$

$$(p \vee \neg q) \equiv (p \vee \neg q) \vee (r \wedge \neg r) \equiv (p \vee \neg q \vee r) \wedge (p \vee \neg q \vee \neg r)$$

$$\therefore PCNF : (p \vee q \vee r) \wedge (p \vee \neg q \vee r) \wedge (p \vee q \vee \neg r) \wedge (p \vee \neg q \vee \neg r)$$

$$Q. \quad \neg (p \rightarrow (q \wedge r))$$

$$\neg (\neg p \vee (q \wedge r))$$

$$p \wedge (\neg q \vee \neg r)$$

$$(p) \wedge (\neg q \vee \neg r)$$

$$\therefore p \equiv p \vee (q \wedge \neg q) \equiv (p \vee q) \wedge (p \vee \neg q)$$

$$p \vee q \equiv (p \vee q) \vee (r \wedge \neg r) \equiv (p \vee q \vee r) \wedge (p \vee q \vee \neg r)$$

$$(p \vee \neg q) \equiv (p \vee \neg q) \vee (r \wedge \neg r) \equiv (p \vee \neg q \vee r) \wedge (p \vee \neg q \vee \neg r)$$

$$(\neg q \vee \neg r) \equiv (\neg q \vee \neg r) \vee (p \wedge \neg p) \equiv (p \vee \neg q \vee \neg r) \wedge (\neg p \vee \neg q \vee \neg r)$$

$$\therefore PCNF : (p \vee q \vee r) \wedge (p \vee \neg q \vee r) \wedge (p \vee q \vee \neg r) \wedge (p \vee \neg q \vee \neg r) \wedge (\neg p \vee \neg q \vee \neg r)$$

Conversion of PDNF to PCNF and vice versa

Method 1:

$$PCNF \rightarrow PDNF$$

1. Let the given expression be A.

2. Then A' will contain all maxterms that are not in A

3. Take (A')' and use DeMorgan Law on RHS

4. The obtained expression is in PDNF.

$$PDNF \rightarrow PCNF$$

$$\text{Min term: } \neg p \rightarrow 0$$

$$\text{Max term: } \neg p \rightarrow 1$$

1. Let the given expression be A.

2. Then A' will contain all min terms not in A.

3. Find $(A')'$ using De Morgan's Law.
 4. Resultant is PCNF.

$$\underline{Q.} \quad (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \vee (p \wedge q \wedge \neg r)$$

$$= \sum (1, 2, 5, 7)$$

$$PCNF = \prod (0, 3, 4, 6)$$

$$= M_0 \wedge M_3 \wedge M_4 \wedge M_6$$

$$= (p \vee q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge (\neg p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$$

Ans.

Q. Convert the given PCNF to PDNF

$$PCNF = (p \vee \neg q \vee r) \wedge (\neg p \vee \neg q \vee \neg r) \wedge (p \vee \neg q \vee \neg r)$$

↓

M_2

↓

M_7

↓

M_3

$$PCNF = \prod (2, 3, 7)$$

$$\therefore PDNF = \sum (0, 1, 4, 5, 6)$$

$$PDNF = m_0 \vee m_1 \vee m_4 \vee m_5 \vee m_6$$

$$= (\neg p \wedge \neg q \wedge \neg r) \vee (\neg p \vee \neg q \vee r) \wedge (p \vee \neg q \vee \neg r) \vee (p \wedge \neg q \wedge r) \vee (p \wedge q \wedge \neg r)$$

Q. $(\neg p \rightarrow r) \wedge (p \Rightarrow q)$

$$(p \vee r) \wedge (p \rightarrow q) \wedge (q \rightarrow p)$$

$$(p \vee r) \wedge (\neg p \vee q) \wedge (\neg q \vee p)$$

$$(p \vee q \vee r) \wedge (p \vee \neg q \vee r) \wedge (\neg p \vee q \vee r) \wedge (\neg p \vee q \vee \neg r) \wedge (p \vee \neg q \vee \neg r)$$

$M_0 \downarrow$

$M_2 \downarrow$

M_4

M_5

M_3

$$\equiv \pi(0, 2, 3, 4, 5)$$

∴ $PDNF \equiv \sum(1, 6, 7) \equiv (\neg p \wedge \neg q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r)$

Ans.

Q. Find PCNF in 3 variables p, q, r

$$(\neg p \wedge r) \vee (p \wedge \neg r)$$

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$$(\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge \neg r)$$

$$PDNF = m_3 \vee m_1 \vee m_6 \vee m_4$$

$$\therefore PCNF \equiv M_0 \wedge M_2 \wedge M_5 \wedge M_7$$

$$\equiv (p \vee q \vee r) \wedge (p \vee \neg q \vee r) \wedge (\neg p \vee q \vee \neg r) \wedge (\neg p \vee \neg q \vee \neg r)$$

Ans.

$$\Phi \quad (p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \vee (p \wedge q \wedge r)$$

$$A \equiv (p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r)$$

$$\neg A \equiv (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r) \vee (p \wedge \neg q \wedge \neg r)$$

$$\neg(\neg A) \equiv (\neg p \vee \neg q \vee r) \wedge (\neg p \vee \neg q \vee \neg r) \wedge (\neg p \vee q \vee r) \wedge (\neg p \vee q \vee \neg r)$$

$\equiv PCNF$ Ans

$$Q_{11}. \quad (p \wedge q) \vee (p \rightarrow r)$$

$$A \equiv (p \wedge q) \vee (\neg p \vee r)$$

$$A \equiv (p \wedge q) \vee (p \wedge \neg r)$$

$$A \equiv \underset{m_7}{(p \wedge q \wedge r)} \vee \underset{m_6}{(p \wedge q \wedge \neg r)} \vee \underset{m_4}{(p \wedge \neg q \wedge \neg r)}$$

$$\Rightarrow \neg A \equiv (\neg p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (p \wedge \neg q \wedge \neg r)$$

$$PCNF: \quad \neg(\neg A) \equiv (p \vee q \vee r) \wedge (p \vee q \vee \neg r) \wedge (\neg p \vee \neg q \vee r) \wedge (\neg p \vee \neg q \vee \neg r) \wedge (\neg p \vee q \vee r)$$

Ans.

Q₁₂. Find PDNF of

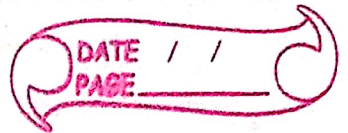
$$(p \vee q) \wedge (\neg p \vee r)$$

$$\equiv \underset{M_0}{(p \vee q \vee r)} \wedge \underset{M_1}{(p \vee q \vee \neg r)} \wedge \underset{M_4}{(\neg p \vee q \vee r)} \wedge \underset{M_6}{(\neg p \vee q \vee \neg r)}$$

$$\therefore PDNF \equiv m_2 \vee m_3 \vee m_5 \vee m_7$$

$$\equiv (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \quad \text{Ans.}$$

Q. $(p \rightarrow q) \wedge (\neg p \wedge q)$
 $(\neg p \vee q) \wedge (\neg p \wedge q) \rightarrow \alpha$



$\therefore$ Using distributive law, we get
 $(\neg p \vee q) \wedge (\neg p \wedge (\neg p \wedge q)) \vee (q \wedge (\neg p \wedge q))$
 $(\neg p \wedge q) \vee (\neg p \wedge q)$
 $\neg p \wedge q$

Q. $p \Rightarrow q$
 $\equiv (p \rightarrow q) \wedge (q \rightarrow p)$
 $(\neg p \vee q) \wedge (\neg q \vee p)$
 $\rightarrow \alpha$

$\therefore (\neg p \wedge (\neg q \vee p)) \vee (q \wedge (\neg q \vee p))$
 $\equiv (\neg p \wedge \neg q) \vee F \vee F \vee (q \wedge p)$
 $\equiv (\neg p \wedge \neg q) \vee (p \wedge q)$ Ans.

Q. Find PCNF of

(i) $\neg(p \vee q)$

$A \equiv \neg p \wedge \neg q$

$\neg A \equiv (p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q)$

$\neg(\neg A) \equiv (\neg p \vee \neg q) \wedge (\neg p \vee q) \wedge (p \vee \neg q)$ Ans.

Q $\neg(p \vee q) \rightarrow (p \wedge q)$

$\equiv (p \vee q) \vee (p \wedge q)$

$\equiv \neg \neg (p \vee q \vee p) \wedge (p \vee q \vee q)$

$\equiv (p \vee q) \wedge (p \vee q)$

PCNF $\equiv (p \vee q)$ Ans.

Theory of Inference

1. $\neg p \Rightarrow p \rightarrow q$
2. $q \Rightarrow p \rightarrow q$
3. $\neg p, p \vee q \Rightarrow q$ (Disjunctive Syllogism)
4. $p, p \rightarrow q \Rightarrow q$ (Modus ponens)
5. $\neg q, p \rightarrow q \Rightarrow \neg p$ (Modus Tollens)
6. $p \rightarrow q, q \rightarrow r \Rightarrow p \rightarrow r$ (Hypothetical syllogism)
7. $p \rightarrow q \Rightarrow \neg q \rightarrow \neg p$ (Contrapositive)

$H_1, H_2, \dots, H_n \Rightarrow C$
Premises Conclusion

$H_1 \wedge H_2 \dots \wedge H_n \rightarrow C$
 $\hookrightarrow$ Tautology

Q. Determine whether the conclusion C is valid in the following when $H_1, H_2, \dots$ are the premises.

(a) $\neg q, p \rightarrow q \Rightarrow \neg p$
 $H_1 \quad H_2 \quad C$

p	q	$\neg q$	$p \rightarrow q$	$\neg p$
T	T	F	T	F
T	F	T	F	F
F	T	F	T	T
F	F	(T)	(T)	(T) $\rightarrow$ Valid

(b) $\neg p, p \vee q \Rightarrow p \wedge q$

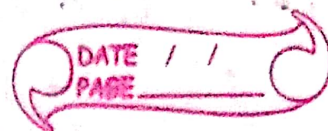
p	q	$\neg p$	$p \vee q$	$p \wedge q$
T	T	F	T	T
T	F	F	T	F
F	T	T	T	F
F	F	T	F	F

$\rightarrow$ invalid

(6)

$$p \rightarrow (q \rightarrow r), \quad p \wedge q \Rightarrow r$$

p	q	r	$p \wedge q$	$p \rightarrow (q \rightarrow r)$
T	T	T	T	T $\rightarrow$ Valid
T	T	F	T	F
T	F	T	F	T
T	F	F	F	T
F	T	T	F	T
F	T	F	F	T
F	F	T	F	T
F	F	F	F	T



Rules of Inference

$\rightarrow$ Rule P: A premise may be introduced at any point in the derivation.

$\rightarrow$ Rule T: A formula S may be introduced in a derivation if S is any one or more of the preceding formulas in the derivation.

Show the validity of the following arguments for which premises are given on left and conclusion on right.

	H_1	H_2	H_3	C
(2)	$\neg(P \wedge \neg Q)$	$\neg Q \vee R$	$\neg R$	$\neg P$

1. $\neg R$ H_3 , Rule P
2. $\neg Q \vee R$ H_2 , Rule P
3. $\neg Q$ 1, 2, disjunctive syllogism
4. $\neg(P \wedge \neg Q)$ H_1 , Rule P
5. $\neg P \vee Q$ 4, De-Morgan's Law
6. $\neg P$ 3, 5, disjunctive syllogism

(b) $(P \wedge Q) \rightarrow R, \neg R \vee S, \neg S \rightarrow \neg P \vee \neg Q$

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- | | | |
|----|------------------------------|-----------------------------|
| 1. | $\neg S$ | H_3 , Rule P |
| 2. | $S \vee \neg R$ | H_2 , Rule P |
| 3. | $\neg R$ | 1, 2, disjunctive syllogism |
| 4. | $(P \wedge Q) \rightarrow R$ | H_1 , Rule P |
| 5. | $\neg(P \wedge Q) \vee R$ | Rule T |
| 6. | $\neg(P \wedge Q)$ | 3, 5 Disjunctive syllogism |
| 7. | $\neg P \vee \neg Q$ | Rule T |

(c) $(P \rightarrow Q) \rightarrow R, P \wedge S, Q \wedge T, R$

- | | | |
|----|-----------------------------------|--------------------|
| 1. | $Q \wedge T$ | H_3 , Rule P |
| 2. | Q | 1, Rule T |
| 3. | $P \wedge S$ | H_2 , Rule P |
| 4. | P | 3, Rule T |
| 5. | $P \rightarrow Q$ | 3, 4, Rule T |
| 6. | $(P \rightarrow Q) \rightarrow R$ | H_1 , Rule P |
| 7. | R | Modus ponens, 5, 6 |

(d) $\neg P \vee Q, \neg Q \vee R, R \rightarrow S, P \rightarrow S$

- | | | |
|----|-------------------|----------------------------|
| 1. | $\neg P \vee Q$ | H_1 , Rule P |
| 2. | $P \rightarrow Q$ | 1, Rule T |
| 3. | $\neg Q \vee R$ | H_2 , Rule P |
| 4. | $Q \rightarrow R$ | 3, Rule T |
| 5. | $P \rightarrow R$ | 2, 4, Hypothesis syllogism |
| 6. | $R \rightarrow S$ | H_3 , Rule P |
| 7. | $P \rightarrow S$ | 5, 6, hypothesis syllogism |

* Rule CP: If we can derive S from R and set of premises then we can derive $R \rightarrow S$ from set of ~~the~~ premises alone.

1. P Rule CP
2. $\neg P \vee Q$ H_1 , Rule P
3. Q 1, 2, disjunctive syllogism
4. $\neg Q \vee R$ H_2 , Rule P
5. R 3, 4, disjunctive syllogism
6. $R \rightarrow S$ H_3 rule
7. S 5, 6 Modus Ponens
8. $P \rightarrow S$ 1, 7, Rule T

Q. P , $P \rightarrow (Q \rightarrow (R \wedge S))$ $Q \rightarrow S$

1. P H_1 , Rule P
2. Q Rule CP
3. $P \rightarrow (Q \rightarrow (R \wedge S))$ H_2 , Rule P
4. $Q \rightarrow (R \wedge S)$ 2, 3 Modus Ponens
5. $R \wedge S$ Modus Ponens
6. R 5, Rule T
7. S 5, Rule T
8. $Q \rightarrow S$ 2, 7, Rule T

Q. H_1 $P \rightarrow (Q \rightarrow R)$, H_2 $Q \rightarrow (R \rightarrow S)$ C $P \rightarrow (Q \rightarrow S)$

1. P Rule CP
2. $P \rightarrow (Q \rightarrow R)$ H_1 , Rule P $Q \rightarrow (R \rightarrow S)$
3. $Q \rightarrow R$ 1, 2, Modus Ponens $\equiv \neg Q \vee \neg R \vee S$
4. $Q \rightarrow (R \rightarrow S)$ Rule P, H_2 $\equiv \neg R \vee \neg Q \vee S$
5. $R \rightarrow (Q \rightarrow S)$ 4, *, Rule T $\equiv R \rightarrow (Q \rightarrow S) *$
6. $Q \rightarrow (Q \rightarrow S)$ 3, 5 hypothetical syllogism
7. $Q \rightarrow S$ 6, **, Rule T
8. $P \rightarrow (Q \rightarrow S)$ 1, 7 Rule T

$$Q. \quad P \rightarrow (Q \rightarrow S), \quad R \rightarrow P, \quad Q \quad \Rightarrow \quad R \rightarrow S$$

1. Q $H_3, \text{Rule P}$
2. R Rule CP

$$3. \quad R \rightarrow P \quad H_2, \text{Rule P}$$

$$4. \quad P \quad 2, 3, \text{Modus ponens}$$

$$5. \quad P \rightarrow (Q \rightarrow S) \quad H_1, \text{Rule P}$$

$$6. \quad Q \rightarrow S \quad 4, 5, \text{Modus ponens}$$

$$7. \quad S \quad 1, 6, \text{Modus ponens}$$

$$8. \quad R \rightarrow S \quad 2, 7, \text{Rule T}$$

Method of Indirect proving / Prove by contradiction

$$Q \quad R \rightarrow \neg Q, \quad R \vee S, \quad S \rightarrow \neg Q, \quad P \rightarrow Q \quad \Rightarrow \quad \neg P$$

$H_1 \quad H_2 \quad H_3 \quad H_4$

$$1. \quad \neg(\neg P) \equiv P \quad (\text{Method of indirect proving})$$

$$2. \quad P \rightarrow Q \quad H_4, \text{Rule P}$$

$$3. \quad Q \quad 1, 2, \text{Modus ponens}$$

$$4. \quad S \rightarrow \neg Q \quad H_3, \text{Rule P}$$

$$5. \quad \neg S \quad 3, 4, \text{Modus tollens}$$

$$6. \quad R \vee S \quad H_2, \text{Rule}$$

$$7. \quad R \quad 5, 6, \text{Rule T}$$

$$8. \quad R \rightarrow \neg Q \quad H_1, \text{Rule P}$$

$$9. \quad \neg Q \quad 7, 8, \text{Modus ponens}$$

$$10. \quad \neg Q \wedge Q \quad 3, 9, \text{Rule T}$$

$$\equiv F$$

Therefore, our assumption that conclusion is false is incorrect.

Hence, conclusion follows from given premises.

Validity of Argument

Test the validity of the following arguments.

1. If two sides of a Δ are equal then opposite angles are equal. Two sides of a Δ are equal. Hence opposite angles are not equal.

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p : Two sides of a Δ are equal
 q : opposite angles are equal

$\therefore$ $p \rightarrow q$, p $\Rightarrow$ $\neg q$
 H_1 H_2

$\therefore$ (1) p H_2 , Rule P

(2) $p \rightarrow q$ H_1 , Rule P

(3) q 1, 2, Modus ponens

$\therefore$ As, we have concluded $q \nRightarrow \neg q$
 $\Rightarrow$ The given argument is invalid.

2. I will become an engineer or mathematician.
 I will not become an engineer.

Therefore, I will become a mathematician

p : I will become an engineer.

q : I will become a mathematician.

$\therefore$ $p \vee q$, $\neg p$ $\Rightarrow$ q
 H_1 H_2 C

(1) $\neg p$ H_2 , Rule P

(2) $p \vee q$ H_1 , Rule P

(3) q 1, 2, ~~Modus tollens~~ Disjunctive syllogism

Hence, the given argument is valid

3. If the Labour market is perfect, then the wages of all persons in a particular employment will be equal. But it is always the case that the wages of such persons are not equal. Therefore the labour

market is not perfect.

p : Labour market is perfect

q : Wages will be equal

$$p \rightarrow q, \neg q \Rightarrow \neg p$$

(1)

$$\neg q$$

H_2 , Rule P

(2)

$$p \rightarrow q$$

H_1 , Rule P

(3)

$$\neg p$$

1, 2, Modus Tollens

Hence, the given argument is valid.

Q. If A works hard then either B or C will enjoy themselves. If B enjoys himself then A will not work hard. If D enjoys himself then C will not. Therefore, if A works hard D will not enjoy himself.

A : A works hard

B : B enjoys

C : C enjoys

D : D enjoys

$\therefore$

$$\begin{array}{ccccccc} A \rightarrow B \vee C & , & B \rightarrow \neg A & , & D \rightarrow \neg C & \Rightarrow & A \rightarrow \neg D \\ H_1 & & H_2 & & H_3 & & C \end{array}$$

1.

A

Rule CP

2

$$B \rightarrow \neg A$$

H_2 , Rule P

3

$$\neg B$$

1, 2, Modus Tollens

4

$$A \rightarrow B \vee C$$

H_1 , Rule P

5

$$B \vee C$$

1, 4, Modus ponens

6

C

3, 5, disjunctive syllogism

7

$$D \rightarrow \neg C$$

H_3 , Rule P

8

$$\neg D$$

6, 7 Modus Tollens

9

$$A \rightarrow \neg D$$

1, 8, Rule T

Hence, given argument is valid.

Q. If Ram misses many classes he fails. If Ram fails then he is uneducated. Ram misses many classes and read lots of books. Hence, he is not uneducated.

p : Ram miss many classes
 q : Ram fails
 r : Ram reads lots of book
 s : Ram is uneducated

$p \rightarrow q, \quad q \rightarrow s, \quad p \wedge r \quad \Rightarrow \quad \neg s$
 $H_1 \quad H_2 \quad H_3 \quad C$

1	$p \wedge r$	$H_3, \text{Rule P}$
2	p	1, Rule T
3	$p \rightarrow q$	$H_1, \text{Rule P}$
4	q	2, 3, Modus ponens
5	$q \rightarrow s$	$H_2, \text{Rule P}$
6	s	4, 5, Modus ponens

As, $s \neq \neg s$

Hence, the given argument is invalid.

Q. If A wins the game then B will be happy. If C wins the game D will be happy. Either A or C will win. However, if A wins D will not be happy and if C wins B will not be happy. So B will be happy if and only if D is not happy.

A : A wins the game
 B : B will be happy
 C : C win the game
 D : D will be happy

$A \rightarrow B, C \rightarrow D, A \vee C, A \rightarrow \neg D, C \rightarrow \neg B \Rightarrow B \leftrightarrow D$
 $H_1 \quad H_2 \quad H_3 \quad H_4 \quad H_5$

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 (B → D) ∧ (D → B)

1	$C \rightarrow \neg B$	H_5 , Rule P
2	$B \rightarrow \neg C$	1, contrapositive
3	$A \vee C$	H_3 , Rule P
4	$\neg C \rightarrow A$	3, Rule T ($C \vee A \equiv \neg C \rightarrow A$)
5	$B \rightarrow A$	2, 4, Hypothetical syllogism
6	$A \rightarrow \neg D$	H_4 , Rule P
7	$B \rightarrow \neg D$	5, 6, Hypothetical syllogism